Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 417: Compiler Design

Batch: 21st | Semester: Fall 2024

Final Examination

Date: 23/03/2025 | Time: 2 hours | Total Marks: 40

Answer any five questions. **Marks: 5 x 8 = 40**

1.	(a)	Derive the summary of a predictive parsing of a compiler.	2
	(b)	Write the pseudocode for LL(1) parsing algorithm of a compiler.	3
	(c)	State the summary of a recursive descent parsing of a compiler.	3
2.	(a)	Where does the reduction happen of a bottom-up parsing algorithm?	2
	(b)	State the conflicts and the sources of conflicts of bottom-up parsing of a compiler.	3
	(c)	Briefly explain the revisited of precedence declarations of a compiler.	3
3.	(a)	State the requirements of LR parsing of a compiler.	2
	(b)	Write the pseudocode for action table of a compiler.	3
	(c)	Draw the pseudocode for producing a set of states of a compiler.	3
4.	(a)	State the differences between abstract syntax tree (AST) and parse tree.	2
	(b)	What does a compiler do to translate an input string?	3
	(c)	Write the Pseudocode for AST node.	3
5.	(a)	What is meant by **scoping** and **scoping overloading**?	2
	(b)	Write the general rules of scoping.	3
	(c)	Compare between **static scoping** and **dynamic scoping**.	3
6.	(a)	Write the general goals of a compiler.	1
	(b)	Why stack machine use in a compiler? Explain.	3
	(c)	Briefly explain the optimization of stack machine and accumulator with stack machine of a compiler.	4
7.	(a)	Derive the epsilon-NFA for the regular expression: ((a*b*)*|(a*+b*)*)*.	2
	(b)	Write the equivalence and minimize DFA algorithm.	6

shift $\Rightarrow \alpha$.

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HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 415: Communication Engineering

Batch: 21st
Date: 20/03/2025

Final Examination

Semester: Fall 2024
Time: 2 Hours
Total Marks: 40

Answer any **five** questions: $5 \times 8 = 40$

1. (a) Describe the point of presence technique in Inter LATA Services. 4
   (b) What are the functions of a mobile switching center? 4

2. (a) List the tasks performed by the signaling system in a Telephone Network. 4
   (b) Describe the SS7 services and its relation to the telephone network. 4

3. (a) In ATM, what is the relationship between TPs, VPs and VCs? 4
   (b) Draw and explain Wireless LAN (IEEE 80.2.11) architecture. 4

4. (a) Explain why SNR in a wireless LAN is normally lower than SNR in a wired LAN? 4
   (b) Compare Piconet with Scatternet in the Bluetooth architecture. 4

5. (a) What is a Footprint? Explain the purposes of GPS. 4
   (b) List the objectives of 4G cellular telephony. 4

6. (a) Write the relationship between the VAN Allen belts with Satellites. 4
   (b) Describe the function of the CDMA in IS-95. 4

7. (a) Distinguish between fixed WiMax and mobile WiMax. 4
   (b) Illustrate a simple network using SONET equipment. 4

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE471: Computer and Network Security

Batch: 21st | Semester: Fall 2024

Date: 15/03/2025 | **Final Examination** | Time: 120 Minutes

Total Marks: 40

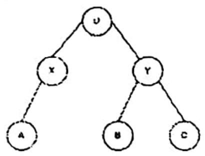
Answer any **five** questions of the following. $5 \times 8 = 40$

1. a) Why MD5 is considered as a broken hashing algorithm? 2
   b) Define non-repudiation and authentication. How do they relate to each other? 3
   c) How does ensuring data integrity contribute to authentication? Explain. 3

2. a) State the differences between hash functions and MAC functions. 3
   b) Describe how a Message Authentication Code (MAC) works. Also describe how MAC ensures both authentication and confidentiality. Provide necessary illustration. 5

3. a) What is the primary purpose of the Oakley Key Determination Protocol, and how does it enhance the Diffie-Hellman key exchange? 4
   b) What is SSL, and what is its primary purpose? Explain the SSL handshaking protocol and illustrate the key steps involved in establishing a secure connection. 4

4. a) What is a digital certificate? 1
   b) Briefly explain the concept of Public Key Infrastructure (PKI) and mention its components. 3
   c) In an X.509 authentication framework, entities U, X, Y, A, B, and C follow the chain of certificates rule. Given the certificate hierarchy as shown, explain how entity A can securely communicate with entity C? 4



5. a) Explain the use of rainbow table in Birthday Attack. 2
   b) What is cryptographic salt? Explain why it is used in cryptographic applications. 2
   c) Explain how PGP (Pretty Good Privacy) works, including its cryptographic processes. Provide two distinct scenarios where PGP is used, and include necessary diagrams or illustrations to support your explanation. 4

6. a) Explain the role of cryptography in the UNIX password scheme, including how cryptographic techniques are used to securely store and verify user passwords. 4
   b) Explain why MIME and S/MIME protocols was introduced after SMTP. 4

7. Illustrate and explain the steps involved in the Kerberos authentication protocol when a client requests access to a local server within a network. 8

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 413 : Computer Graphics

Batch: 21st | Semester: Fall 2024

Time: 2 Hours

Date: 17/03/2025 | **Final Examination** | Total Marks: 40

Answer any **five** questions. $5 \times 8 = 40$

1. (a) Define Window and Viewport. Illustrate the two-dimensional viewing transformation in computer graphics. **4**

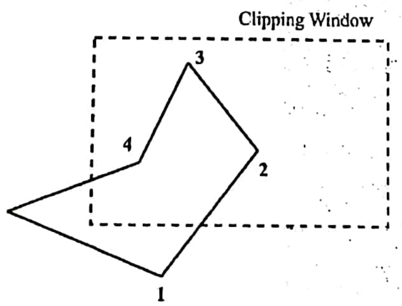
   (b) Express the window-to-viewport mapping in the form of a composite transformation matrix. Find the normalization transformation that maps a window whose lower left corner is at (2,1) and upper right corner is at (4,5) onto a viewport that is the entire normalized device screen. **4**

2. (a) Correctly label and show the region codes used in the Cohen-Sutherland line clipping algorithm based on their positions relative to the clipping window. **2**

   (b) Consider a rectangular clipping window with corners at (-3,1) and (2,6). Three line segments have endpoints: A(-1, 5), B(3, 8); C(-4, 7), D(-2, 10) and E(-2, 3), F(1, 2). Now, using the Cohen-Sutherland algorithm, determine if the line segments are either visible, or partially visible, or invisible. For partially visible segments, find the intersection points with the window boundaries. **6**

3. (a) Explain the limitations of line clipping algorithms when applied to polygon fill-area clipping. **3**

   (b) Clip the polygon with vertexes 1,...,5 in the following figure against the rectangular clipping window using the Sutherland-Hodgman algorithm. **5**



4. (a) State the transformation matrices for three-dimensional reflection in the xy, yz and zx plane. **3**

   (b) Given a homogeneous point (3, 1, 2), apply 60 degree rotation towards X, Y and Z axis and find out the new coordinate points. **5**

5. (a) Briefly explain the steps involved in the three-dimensional viewing pipeline. **4**

   (b) Define projection in computer graphics. Differentiate between parallel and **4**

X → y → Z → x

perspective projection with necessary figures.

6. (a) Define hue and saturation. Explain the CMY color model with a figure to illustrate the combinations of its primary colors. 4

(b) Convert an RGB color with values R = 102, G = 153, and B = 204 to the YIQ color space using the appropriate transformation formulas. 4

7. (a) Define vanishing point in projection. Illustrate different types of perspective projection based on the number of principal vanishing points. 3

(b) State the steps for Cohen-Sutherland line clipping algorithm. 5

C -> cyan -> Red absorbs

M -> Magenta -> Green

Y -> Yellow -> Blue

hue -> attribute

saturation -> purity -> intensity or vividness

A space in world coor to view objects

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 411: Artificial Intelligence

Batch: 21st | Semester: Fall 2024

Time: 2 hours

Date: 12/03/2025 | **Final Examination** | Total Marks: 40

Answer any **five** questions from the following: 5x8= 40

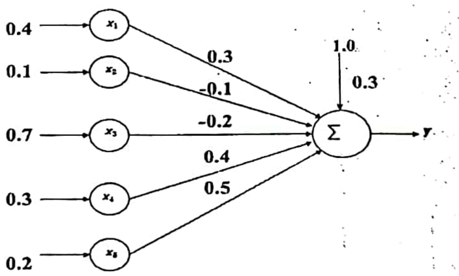
1 (a) List the various distance measures used in clustering. 2

(b) Apply agglomerative clustering to the following unlabeled dataset and present the final results using a dendrogram. 6

	A	B	C	D	E	F
A	0					
B	520	0				
C	815	410	0			
D	300	720	610	0		
E	650	365	290	520	0	
F	900	580	455	775	440	0

2 (a) Draw the block diagram of multi-layer neural network. 2

(b) Find the output of the following net when (i) Step function (θ =0), (ii) Step function (θ =0.5), (iii) Sign function (θ =-0.5), (iv) Sign function (θ =1.0), (v) Binary sigmoid function, and (vi) Bipolar sigmoid function are used. 6



3 (a) Which metrics are used to measure the performance of a classification method? 2

(b) Determine the most suitable splitting attribute by evaluating the Gini Index of each attribute using the following dataset. 6

Outlook	Humidity	PlayTennis
Sunny	High	No
Sunny	Normal	Yes
Overcast	High	Yes
Rain	High	Yes
Rain	Normal	No

4 (a) Outline the types of algorithms utilized for tree induction in decision trees. 2

(b) Using the Naive Bayes model, classify a new sample with the feature values Age: 34, Income: 55000, and Education Level: Bachelor's to predict whether the person will buy a product based on the following dataset. 6

Age	Income	Education Level	Buy Product
25	30000	High School	No
34	45000	Bachelor's	Yes
34	55000	Master's	Yes
40	70000	PhD	Yes
22	25000	High School	No
28	40000	Bachelor's	Yes
34	60000	Master's	Yes
45	80000	PhD	Yes

106.25

210

5 (a) Explain the disadvantages of Naïve Bayes classifier. 3

(b) Predict the exam score for a new sample where the student studied for 9 hours using Linear Regression model based on the following dataset. 5

Hours Studied (X)	Exam Score (Y)
1	50
2	55
3	60
4	65
5	70
6	75
7	80
8	85

$w_0 = \bar{y} - w_1 \bar{x}$

$y = w_0 + w_1 x$

6 (a) Imagine you have the following document: "The fast red fox leaps over the sleepy cat. The red fox is extremely fast." Using a Count Vectorizer, demonstrate how you would convert this sentence into a numerical representation. 3

(b) Briefly outline the stages of the NLP pipeline and how they work together. 5

7 (a) Explain the analogy between biological and artificial NNs. 4

(b) State the differences between supervised and unsupervised learning. 4